

Questionnaire 2025: Solutions

Shine Jose

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Solution 1

Evaluate the given expression,[1]

$$\lim_{x \rightarrow a} \frac{x^x - x^a}{a^x - a^a}$$

Step 1: Recognise The Given Form

As $x \rightarrow a$, both the numerator and the denominator approach 0, [2]

$$x^x \rightarrow x^a, \quad a^x \rightarrow a^a \Rightarrow \text{Indeterminate form } \frac{0}{0}$$

Step 2: Use Logarithmic Representation

Recall, [3]

$$x^x = e^{x \log x} \Rightarrow \text{differentiable for } x > 0$$

Step 3: Differentiate Numerator & Denominator

Let, [4]

$$\begin{aligned} f(x) = x^x &\Rightarrow f'(x) = x^x(\log x + 1) \\ g(x) = a^x &\Rightarrow g'(x) = a^x \log a \end{aligned}$$

Step 4: Apply L'Hôpital's Rule

Using L'Hôpital's rule, [4]

L'Hôpital's rule is a powerful tool in calculus used to evaluate limits of indeterminate forms such as $\frac{0}{0}$ or $\frac{\infty}{\infty}$. It states that if the limit of a function divided by another results in an indeterminate form, you can find the limit by taking the derivatives of both the numerator and the denominator and then evaluating the limit of the new fraction.

$$\lim_{x \rightarrow a} \frac{x^x - x^a}{a^x - a^a} = \frac{f'(a)}{g'(a)} = \frac{a^a(\log a + 1)}{a^a \log a} = \boxed{\frac{\log a + 1}{\log a}}$$

Step 5: Interpret The Result

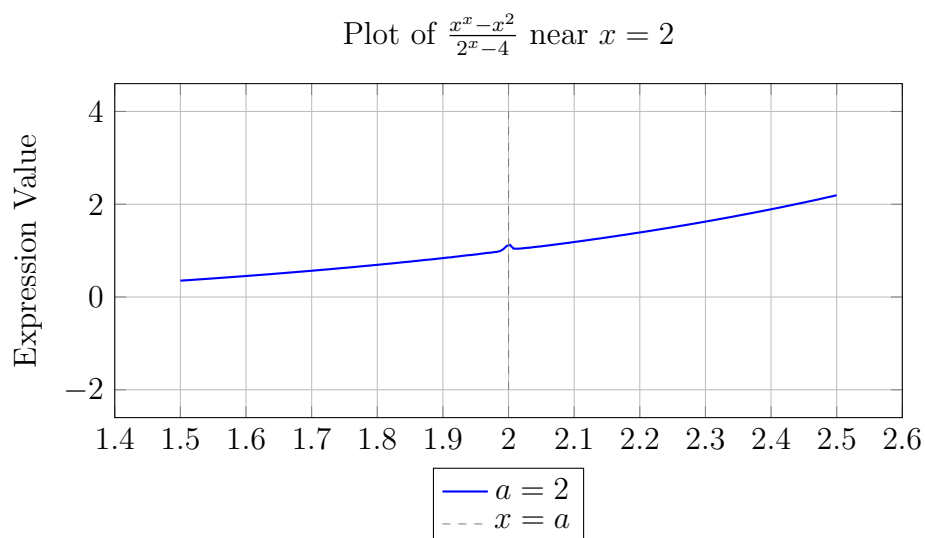
This expression is undefined at $a = 1$ because $\log 1 = 0$, so we must avoid that point.

Step 6: Visualize The Behaviour

Below is a plot of the expression for $a = 2$,

Graphical Representation

Below is a plot of the limit expression $\frac{x^x - x^a}{a^x - a^a}$ around $x = a$ with $a = 2$, [4]



Solution 2

An orthonormal basis in $\mathbb{R}^n$ is a set of n vectors that are,

- Unit vectors: $\|v_i\| = 1$
- Mutually orthogonal: $v_i \cdot v_j = 0$ for $i \neq j$

vectors Present

There are n vectors in the basis because the dimension of $\mathbb{R}^n$ is n ,

Sets A and B

If A and B are disjoint subsets of the orthonormal basis,[5]

- The vectors in A cannot be linearly dependent on any vector in B because the orthonormal vectors are all linearly independent.
- Therefore, no vector in A lies in the span of B . [6]

Matrix P and $P^\top P$

Let A be an $n \times k$ matrix of orthonormal columns (with $k \leq n$) and 0 be an $n \times (n - k)$ zero matrix[7] [8].

$$P = [A \ 0] \Rightarrow P^\top P = \begin{bmatrix} A^\top A & 0 \\ 0 & 0 \end{bmatrix}$$

Since A has orthonormal columns:

$$A^\top A = I_k \Rightarrow P^\top P = \begin{bmatrix} I_k & 0 \\ 0 & 0 \end{bmatrix}$$

Section 3

Given scalar-valued function,

$$E[u] = \sum_{i=1}^1 \sum_{j=0}^1 \cos(u[i, j]) \sin(u[i-1, j])$$

Step 1: Understand Indexing

Since $u \in \mathbb{R}^{2 \times 2}$ and $i = 1, j = 0, 1$, substituting

$$E[u] = \cos(u[1, 0]) \sin(u[0, 0]) + \cos(u[1, 1]) \sin(u[0, 1])$$

Step 2: Label Variables for Calculation

Let,

$$a = u[1, 0], \quad b = u[0, 0], \quad c = u[1, 1], \quad d = u[0, 1]$$
$$E[u] = \cos(a) \sin(b) + \cos(c) \sin(d)$$

Step 3: Compute Partial Derivatives

Using the chain rule,

$$\frac{\partial E}{\partial u[1, 0]} = -\sin(a) \sin(b)$$
$$\frac{\partial E}{\partial u[0, 0]} = \cos(a) \cos(b)$$
$$\frac{\partial E}{\partial u[1, 1]} = -\sin(c) \sin(d)$$
$$\frac{\partial E}{\partial u[0, 1]} = \cos(c) \cos(d)$$

Step 4: Computing Gradient Matrix

$$\nabla_u E = \begin{bmatrix} \cos(u[1, 0]) \cos(u[0, 0]) & \cos(u[1, 1]) \cos(u[0, 1]) \\ -\sin(u[1, 0]) \sin(u[0, 0]) & -\sin(u[1, 1]) \sin(u[0, 1]) \end{bmatrix}$$

Each element of $\nabla_u E$ corresponds to the derivative of $E[u]$ with respect to the matching index of $u[i, j]$,

Each element of the gradient tells you how sensitive the output $E[u]$ is to a small change in the corresponding component of the matrix $[u]$. This is essential for optimization algorithms like gradient descent when training models with matrix parameters[9][10].

Section 4

Step 1: Probability Mass Function (PMF)

The probability mass function of the Poisson distribution is[11][12],

$$p(x; \lambda) = \frac{e^{-\lambda} \lambda^x}{x!}, \quad x \in \{0, 1, 2, \dots\}, \quad \lambda > 0$$

Where:

- λ is the rate (mean number of events in a fixed interval).
- x is the number of events observed.

Step 2: Expected Value $\mathbb{E}[x]$

We compute the expectation of the Poisson distribution:

$$\mathbb{E}[x] = \sum_{x=0}^{\infty} x \cdot \frac{e^{-\lambda} \lambda^x}{x!} = e^{-\lambda} \sum_{x=1}^{\infty} \frac{x \lambda^x}{x!} = e^{-\lambda} \sum_{x=1}^{\infty} \frac{\lambda^x}{(x-1)!}$$

Change the index to $y = x - 1$:

$$\mathbb{E}[x] = e^{-\lambda} \cdot \lambda \sum_{y=0}^{\infty} \frac{\lambda^y}{y!} = \lambda \cdot e^{-\lambda} \cdot e^{\lambda} = \boxed{\lambda}$$

Step 3: Maximum Likelihood Estimate (MLE) of λ

Let $x_1, x_2, \dots, x_m$ be i.i.d. samples from a Poisson distribution. The likelihood is

$$L(\lambda) = \prod_{i=1}^m \frac{e^{-\lambda} \lambda^{x_i}}{x_i!} = e^{-m\lambda} \lambda^{\sum x_i} \prod_{i=1}^m \frac{1}{x_i!}$$

Taking the logarithm,

$$\log L(\lambda) = -m\lambda + \left(\sum_{i=1}^m x_i \right) \log \lambda - \sum_{i=1}^m \log x_i!$$

Differentiate concerning λ ,

$$\frac{d}{d\lambda} \log L(\lambda) = -m + \frac{1}{\lambda} \sum_{i=1}^m x_i = 0 \Rightarrow \boxed{\hat{\lambda} = \frac{1}{m} \sum_{i=1}^m x_i}$$

So, the MLE of λ is the sample mean.

[12] [11] [1] [13] [7] [4] [10] [8] [5] [3] [2] [9] [6]

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